



LIGHT ANIMATIONS

Using Arduino And MATLAB

SHIBENDU MAHATA AND SAIKAT PATRA

Light animations are visually appealing and hence widely used for advertising purposes. In this project, we present a MATLAB-based graphical user interface (GUI) approach to control the glowing pattern of a number of light-emitting diodes (LEDs). Use of GUI is advantageous since the user can control illumination patterns while performing other tasks in the PC.

This project creates five different lighting patterns including ring counter and Johnson counter by clicking appropriate pushbuttons in the GUI. The blinking speed of LEDs can also be controlled using fast, normal and slow pushbuttons in the GUI. The author's prototype is shown in Fig. 1 and the MATLAB-based GUI in Fig. 2.

Circuit and working

The circuit for controlling light animations using Arduino is shown in Fig. 3. It consists of an Arduino Uno board, eight LEDs and eight 1-kilo-ohm resistors.

Arduino Uno is an AVR ATmega328P microcontroller based development board with six analogue input pins and 14 digital input/output (I/O) pins. The microcontroller has 32kB of ISP flash memory, 2kB RAM and 1kB EEPROM. The board provides serial

communication via UART, SPI and I2C. The microcontroller can operate at a clock frequency of 16MHz. In this project, digital I/O pins 3 through 10 of

the Arduino are configured as output pins and used to control the illumination of eight LEDs.

When a pushbutton corresponding to a particular pattern is pressed in the GUI, it executes the corresponding callback function in the source program (Illumination001.m.). The callback function executes program statements corresponding to that pattern and sends High or Low control signals to appropriate pins of the Arduino in order to create the desired glowing pattern. Eight series-connected 1-kilo-ohm resistors limit current flow through LEDs.

Software

We used Arduino IDE to program the Arduino Uno. The latest IDE can be downloaded for free from Arduino's official website. After download, install the Arduino IDE and note the installation directory.

Download 'Legacy MATLAB and Simulink Support for Arduino' package from Mathworks website. Extract the compressed folder named ArduinoIO. From ArduinoIO folder, copy pde folder and paste it in C:\Program Files(or Program files x86)\Arduino\libraries. (The path may be different depending on the installation directory of the IDE). After pasting pde folder in the

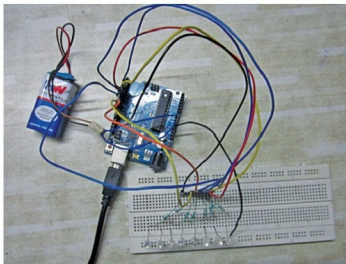


Fig. 1: Author's prototype of the project



Fig. 2: MATLAB-based GUI